

HANO, SwedenWMO: **026280**Lat: **56.0137N**Lon: **14.8462E**Elev: **197**StdP: **14.59**Time Zone: **1.00 (EUC)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	19.1	22.0	12.0	10.2	22.7	15.2	11.9	24.5	44.0	36.5	41.0	35.5	17.9	40	0.812

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.2	74.6	65.7	72.1	64.8	69.9	63.5	67.8	72.0	66.1	70.4	64.6	68.7	11.9	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.1	96.9	70.6	64.5	91.5	68.6	62.9	86.3	67.2	32.3	72.2	31.0	70.6	29.8	68.7	75.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 38.8	34.5	31.2	DB	16.1	79.5	3.9	3.4	13.2	81.9	11.0	83.9	8.8	85.8	5.9	88.2	(4)
(5)			WB	14.9	69.0	3.9	2.2	12.1	70.6	9.8	71.9	7.6	73.1	4.7	74.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	47.1	33.4	33.3	36.4	42.4	51.0	58.0	63.2	63.0	56.8	48.6	41.8	36.3	(6)	
	DBStd	11.87	5.74	5.48	5.08	4.37	4.71	4.29	4.56	4.26	4.14	4.79	5.10	6.10	(7)	
	HDD50	2433	515	467	422	230	43	1	0	0	1	82	248	424	(8)	
	HDD65	6605	980	887	887	678	435	214	92	91	247	508	697	889	(9)	
	CDD50	1373	0	0	0	3	73	241	409	402	205	39	1	0	(10)	
	CDD65	69	0	0	0	0	0	4	36	28	1	0	0	0	(11)	
	CDH74	109	0	0	0	0	0	8	71	30	0	0	0	0	(12)	
	CDH80	5	0	0	0	0	0	0	3	2	0	0	0	0	(13)	
(14) Wind	WSAvg	16.3	18.7	18.4	17.0	15.2	14.5	13.7	12.9	14.0	16.0	17.6	18.4	19.1	(14)	
(15) Precipitation	PrecAvg	23.4	1.8	1.4	1.1	1.3	1.7	2.2	2.5	2.3	2.2	2.4	2.5	2.0	(15)	
	PrecMax	31.1	3.2	3.0	2.6	2.5	5.7	6.1	8.7	5.7	6.6	5.0	4.8	4.2	(16)	
	PrecMin	17.8	0.0	0.4	0.2	0.1	0.3	0.0	0.2	0.2	0.4	0.4	0.6	0.8	(17)	
	PrecStd	3.2	0.9	0.7	0.6	0.7	1.0	1.3	1.6	1.3	1.5	1.2	1.0	0.7	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.9	46.7	52.5	57.9	67.7	74.7	79.5	77.3	70.2	59.7	53.9	48.6	(19)	
		MCWB	44.3	43.5	45.4	49.4	57.0	62.9	66.1	67.4	64.3	56.7	51.8	46.5	(20)	
	2%	DB	43.5	43.5	47.4	54.1	63.6	69.9	75.9	73.8	67.1	57.5	51.5	46.9	(21)	
		MCWB	41.9	41.4	42.1	47.2	55.9	60.7	66.2	66.0	61.3	55.0	49.0	45.1	(22)	
	5%	DB	41.9	41.7	44.7	51.5	60.7	67.2	73.0	71.6	64.6	56.1	49.7	45.4	(23)	
		MCWB	40.5	39.8	40.8	46.3	54.2	59.4	65.2	65.2	59.8	53.8	47.3	43.8	(24)	
	10%	DB	40.4	40.1	42.6	49.1	58.4	64.9	70.6	69.4	62.6	54.8	48.1	43.8	(25)	
		MCWB	39.1	38.5	39.8	44.5	52.5	58.4	64.2	63.9	58.5	52.7	46.4	42.6	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.3	44.5	46.9	51.4	60.0	65.3	70.2	69.6	66.2	58.0	52.7	47.4	(27)	
		MCDB	45.2	45.7	51.4	56.3	64.3	71.0	74.8	74.5	69.8	58.9	53.4	48.3	(28)	
	2%	WB	42.5	41.9	43.2	48.8	57.4	62.5	68.4	67.5	62.7	55.8	50.0	45.5	(29)	
		MCDB	43.3	42.9	45.7	52.7	61.6	67.2	72.8	71.6	65.5	56.9	51.1	46.5	(30)	
	5%	WB	40.8	40.1	41.5	46.7	55.3	60.9	66.5	66.2	60.9	54.4	48.1	44.2	(31)	
		MCDB	41.6	41.3	43.8	50.4	59.8	65.3	71.1	70.1	63.5	55.6	49.3	45.2	(32)	
	10%	WB	39.3	38.7	40.1	45.0	53.4	59.4	65.0	64.8	59.4	53.1	46.7	42.7	(33)	
		MCDB	40.3	39.9	42.2	48.4	57.4	63.7	69.5	68.3	61.7	54.4	47.8	43.7	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	4.9	5.3	7.3	8.7	9.6	10.0	10.2	8.9	7.7	5.8	4.7	4.8	(35)	
		MCDBR	4.9	6.6	10.9	12.5	12.9	13.5	13.0	10.7	9.7	6.4	4.9	5.1	(36)	
	5% WB	MCWBR	4.9	5.4	7.2	7.5	7.3	7.3	6.6	6.0	5.7	5.0	5.0	5.1	(37)	
		MCWBR	4.6	6.2	9.8	11.5	11.2	11.4	10.9	9.2	8.0	5.7	4.8	5.0	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.309	0.334	0.353	0.386	0.384	0.376	0.399	0.400	0.369	0.356	0.336	0.301	(40)	
		taud	2.292	2.322	2.334	2.313	2.375	2.427	2.381	2.381	2.446	2.403	2.332	2.266	(41)	
	Ebn at Noon		188	230	256	264	272	275	266	257	250	220	175	159	(42)	
		Edh at Noon	17	24	30	35	35	34	35	33	27	22	16	14	(43)	
(44) All-Sky Solar Radiation	RadAvg		131	325	779	1324	1740	1855	1694	1367	954	450	172	93	(44)	
	RadStd		25	59	75	150	171	119	196	114	105	78	32	10	(45)	

Historical Trends

		Heating			Cooling		
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